THREE CONSTRUCTIONS OF THE QUASI-COHERENT SHEAF ASSOCIATED TO A MODULE OVER THE RING OF GLOBAL SECTIONS AGREE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The pullback of modules along the structure morphism to the point with the given ring, and its description as a sheafification of a tensor product presheaf. Invoked inside the counted proof (source lines 1402-1405) as a cross-chapter pointer. Identifies the first and the third construction.
- The right exactness of the pullback and its compatibility with direct sums. Invoked inside the counted proof (source lines 1406-1411) as a pointer to a lemma of this chapter. Transports a presentation of the module to a presentation of the pulled back sheaf, which identifies the second construction with the first.

Source reference. Original lemma lemma-exactness-pushforward-pullback: sources/stacks-modules/pinned-original-source.tex, source lines 990–990. Navigation pointer only: the counted text refers to the Stacks item lemma-exactness-pushforward-pullback, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference sheaves-section-ringed-spaces-functoriality-modules: sources/stacks-modules/pinned-original-source.tex, source lines 1405–1405. Navigation pointer only: the counted text refers to the Stacks item sheaves-section-ringed-spaces-functoriality-modules, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Source reference. Original reference sheaves-lemma-stalk-pullback-modules: sources/stacks-modules/pinned-original-source.tex, source lines 1418–1418. Navigation pointer only: the counted text refers to the Stacks item sheaves-lemma-stalk-pullback-modules, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be ringed space. Let $\alpha : R \to \Gamma(X, \mathcal{O}_X)$ be a ring homomorphism from a ring R into the ring of global sections on X. Let M be an R-module. The following three constructions give canonically isomorphic sheaves of $\mathcal{O}_X$-modules:*

1. *Let $\pi : (X, \mathcal{O}_X) \longrightarrow (\{*\}, R)$ be the morphism of ringed spaces with $\pi : X \to \{*\}$ the unique map and with π-map $\pi^\sharp$ the given map $\alpha : R \to \Gamma(X, \mathcal{O}_X)$. Set $\mathcal{F}_1 = \pi^* M$.*
2. *Choose a presentation $\bigoplus_{j \in J} R \to \bigoplus_{i \in I} R \to M \to 0$. Set*
$$\mathcal{F}_2 = \text{Coker}\left(\bigoplus\nolimits_{j \in J} \mathcal{O}_X \to \bigoplus\nolimits_{i \in I} \mathcal{O}_X\right).$$
Here the map on the component $\mathcal{O}_X$ corresponding to $j \in J$ given by the section $\sum_i \alpha(r_{ij})$ where the r_{ij} are the matrix coefficients of the map in the presentation of M.
3. *Set $\mathcal{F}_3$ equal to the sheaf associated to the presheaf $U \mapsto \mathcal{O}_X(U) \otimes_R M$, where the map $R \to \mathcal{O}_X(U)$ is the composition of α and the restriction map $\mathcal{O}_X(X) \to \mathcal{O}_X(U)$.*

This construction has the following properties:

1. *The resulting sheaf of $\mathcal{O}_X$-modules $\mathcal{F}_M = \mathcal{F}_1 = \mathcal{F}_2 = \mathcal{F}_3$ is quasi-coherent.*
2. *The construction gives a functor from the category of R-modules to the category of quasi-coherent sheaves on X which commutes with arbitrary colimits.*
3. *For any $x \in X$ we have $\mathcal{F}_{M,x} = \mathcal{O}_{X,x} \otimes_R M$ functorial in M.*
4. *Given any $\mathcal{O}_X$-module $\mathcal{G}$ we have*
$$\text{Mor}_{\mathcal{O}_X}(\mathcal{F}_M, \mathcal{G}) = \text{Hom}_R(M, \Gamma(X, \mathcal{G}))$$
where the R-module structure on $\Gamma(X, \mathcal{G})$ comes from the $\Gamma(X, \mathcal{O}_X)$-module structure via α.

Proof. The isomorphism between $\mathcal{F}_1$ and $\mathcal{F}_3$ comes from the fact that π^* is defined as the sheafification of the presheaf in (3), see Sheaves, Section sheaves-section-ringed-spaces-functoriality-modules. The isomorphism between the constructions in (2) and (1) comes from the fact that the functor π^* is right exact, so $\pi^*(\bigoplus_{j \in J} R) \to \pi^*(\bigoplus_{i \in I} R) \to \pi^* M \to 0$ is exact, π^* commutes with arbitrary direct sums, see Lemma lemma-exactness-pushforward-pullback, and finally the fact that $\pi^*(R) = \mathcal{O}_X$.

Assertion (1) is clear from construction (2). Assertion (2) is clear since π^* has these properties. Assertion (3) follows from the description of stalks of pullback sheaves, see Sheaves, Lemma sheaves-lemma-stalk-pullback-modules. Assertion (4) follows from adjointness of π_* and π^*. □